REVIEWING RESULTS IN OHDSI SHINY APPS

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WHAT THEY ARE & WHY THEY MATTER?

Shiny Apps are interactive web applications built using R.

- Allow users to interact with data – explore, filter, and visualize data in real time.
- Share results with collaborators.

WHY THEY MATTER?

SHARING RESULTS

STUDY PROTOCOL → PHENOTYPE/CODE CREATION → EXECUTION OF CODE → DISSEMINATION

- Nice addition to publications/reports -> **Much more than an optional, final step!**

WHY THEY MATTER?

- Support multiple stages of study development.
- Help detect inconsistencies and areas needing attention early in the process.
- Promote transparency (by making ALL study results public).

WHY THEY MATTER?

DIAGNOSTICS

- To examine phenotypes
- To examine drug exposures

RESULTS REVISION

- Specific to the study (often with common features)

AGENDA

Time	Activity
15:30-16:25	Diagnostics
16:25-16:35	Break
16:35-17:20	Study results
17:20-17:30	Closure and questions

HANDS-ON REVIEW OF SHINY APPS

Be ready to get hands-on!

DIAGNOSTICS

SHINY APPS FOR DIAGNOSTICS

To examine phenotypes:

- CohortDiagnostics
- PhenotypeR

Based on cohorts:
Set of persons who satisfy one or more inclusion criteria for a duration of time

To examine drug exposures:

- DrugExposureDiagnostics

Based on ingredients
It only needs: Ingredient concept ID

SHINY APPS FOR DIAGNOSTICS

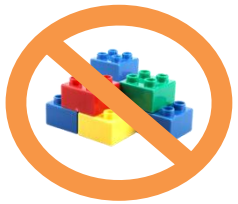
ATHENA

acetaminophen

DETAILS	
Domain ID	Drug
Concept Class ID	Ingredient
Vocabulary ID	RxNorm
Concept ID	1125315
Concept code	161
Validity	Valid
Concept	Standard
Valid start	01-Jan-1970
Valid end	31-Dec-2099

Based on ingredients

It only needs: Ingredient concept ID



https://athena.ohdsi.org/search-terms/terms/1125315

INTRODUCTION

Summarise drug records that contain a specific ingredient based on information recorded in the **drug_exposure table**.

- Concepts (mapped & source)
- Drug routes
- Type of records
- Duration
- Quantity
- Daily dose
- Missing data
- Verbatim instructions

OMOP CDM: https://ohdsi.github.io/CommonDataModel/cdm54.html#drug_exposure
Inberg G, Burn E, Burkard T (2026). *DrugExposureDiagnostics: Diagnostics for OMOP Common Data Model Drug Records*. R package version 1.1.7, https://darwin-eu.github.io/DrugExposureDiagnostics/.

EXAMPLE CASE

DARWIN EU® - Paracetamol prescribing and paracetamol overdose in Europe: a descriptive analysis of trends and patient characteristics

Objectives:

1. To examine the **incidence/prevalence of paracetamol prescribing**.
2. To examine the **incidence of paracetamol overdose.**
3. To **characterise patients with paracetamol overdose**.

Protocol and report available in the HMA- EMA RWD Catalogue: EUPAS1000000329 | EUPAS1000000584

EXAMPLE CASE

- Put yourself in the shoes of the team conducting this multi-database study.
- You have received Diagnostics results from 6 hypothetical databases. Explore them, focusing on those most relevant to the research question.
- Try to anticipate and identify any critical issues that need to be addressed to prevent delays and ensure the validity of the study.

INTRODUCTION

Study Objective: To examine the incidence/prevalence of paracetamol prescribing (overall, by age, sex and formulation)

- Concepts (mapped & source)
- Drug routes
- Type of records
- Duration
- Quantity
- Daily dose
- Missing data
- Verbatim instructions

YOUR TURN

EXAMPLE CASE

- Put yourself in the shoes of the team conducting this multi-database study.
- You have received Diagnostics results from 6 hypothetical databases.

LINKS:

- **Exercise (HTML Part 1):**

https://github.com/Braventos/OHDSI_Symposium26_ShinyApps/

- **Shiny (link also included in the exercise):**

https://data.darwin-eu.org/ShinyWorkshop_DrugExpDiagnostics/

EUROPE

Concept Class	Composed of
Marketed Product	Ingredients, their strength, form, supplier (brand name and box size are optional)
Quantified Branded Drug Box	Ingredients, their strength, form, brand name, size and box size
Quantified Clinical Drug Box	Ingredients, their strength, form, size and box size
Branded Drug Box	Ingredients, their strength, form, brand name and box size
Clinical Drug Box	Ingredients, their strength, form and box size
Quantified Branded Drug	Ingredients, their strength, form, brand name, size
Quantified Clinical Drug	Ingredients, their strength, form, size
Branded Pack	Branded Drugs, their number (box size is optional)
Clinical Pack	Clinical Drugs, their number (box size is optional)
Branded Drug	Ingredients, their strength, form, brand name
Clinical Drug	Ingredient, their strength, form
Branded Drug Form	Ingredient, form, brand name
Clinical Drug Form	Ingredient, form (dose cannot be inferred)
Branded Drug Component	Ingredient, strength, brand name
Clinical Drug Component	Ingredient, strength
Dose Form	Form
Brand Name	Brand name
Ingredient	Ingredient
Drug Class	Drug class

BREAK

STUDY RESULTS

EXAMPLE CASE

Key methodological aspects:

- Design: Cohort study (2010-2023)
- Population:
  - Incidence/prevalence: All individuals with >1y of prior observation
  - Characterisation: Individuals with paracetamol overdose
- Data sources: 7

EXAMPLE CASE

Data sources:

Country	Data Source	Health care setting	Type of data
Denmark	DK-DHR	Hospital	EHR, registry
France	APHM[1]	Hospital	EHR, registry, others
Germany	InGef RDB[1]	Primary care, hospital	Claims
Netherlands	IPCI	Primary care	EHR
Norway	NLHR	Primary care, hospital	EHR
Spain	H12O	Hospital	EHR, registry
Sweden	HI-SPEED[1]	Primary care[2], hospital	Registry

[1] Data availability started later in APHM (2014), InGef RDB (2015) and HI-SPEED (2015; 2018 for drugs).
[2] Primary care data is available for 40% of the population.

STUDY RESULTS

- We will look at one Shiny app with results together
- You will open another Shiny app for hands-on reviewing the results

  Please go to this web page to open the exercises:

  https://data.darwin-eu.org/EUPAS1000000584/

- Time for plenary questions and discussion

CLOSURE AND QUESTIONS